

RAMYA K

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building and owning backend and full-stack systems in large-scale production environments. Experienced in designing and operating distributed microservices using Java, Spring Boot, Kafka, and cloud platforms including AWS, Azure, and GCP, with additional experience developing services using .NET and C#. Proven track record of delivering features end-to-end, improving system latency and throughput, and supporting high-traffic production services through monitoring, on-call support, and incident resolution.

TECHNICAL SKILLS

- **Programming Languages:** Java 8, C#, Python, JavaScript (ES6+), TypeScript, Kotlin, SQL
- **Backend & Microservices:** Spring Core, Spring Boot, Spring Data JPA, Spring Security, Spring WebFlux, Spring AOP, Hibernate, .NET Core, FastAPI, Node.js, GraphQL, Rest APIs, Resilience4J
- **Java Technologies:** JDBC, JPA, JMS, JTA, Servlets, JAX-WS
- **Frontend:** React.js, Angular, Reactive Forms, Angular Material, Svelte, RxJS, HTML5, CSS3
- **Databases:** PostgreSQL, MySQL, MongoDB, Oracle (PL/SQL), DynamoDB, Cassandra
- **Cloud Platforms:** AWS (EC2, VPC, RDS, S3, Lambda, CloudWatch), Azure (Functions, Event Hubs, Blob Storage, Application Insights, Azure Monitor), GCP(GKE, Cloud Run, Pub/Sub, Cloud SQL)
- **DevOps and Build Tools:** Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Azure DevOps, Maven, Git
- **Messaging Systems:** RabbitMQ, Apache Kafka
- **Monitoring & Quality:** Splunk, Zipkin, SonarQube, Nexus Repository
- **Testing:** JUnit, Mockito, Jest, Playwright, Cypress, Selenium WebDriver, Postman, SOAPUI, JMeter
- **AI & LLM Technologies:** Azure AI Document Intelligence, Azure AI Language, Vertex AI, LangChain, Retrieval-Augmented Generation (RAG), OpenAI APIs, Hugging Face Transformers, Pinecone, FAISS, TensorFlow, Keras, Vector Search, LangFlow

PROFESSIONAL EXPERIENCE

State of Ohio | *Software Developer*

May 2026 - Present

- Collaborate with business stakeholders, product owners, analysts, and solution architects using JIRA and Confluence to analyze public-sector workflows, gather requirements, and map integration dependencies, producing technical designs and delivery estimates that accelerated project planning by clarifying scope
- Design and refine responsive user experiences in Figma, applying WCAG accessibility guidelines and creating reusable component specs for desktop and mobile devices, which reduced redesign cycles and improved user satisfaction.
- Develop modular, responsive front-end applications using Angular, Svelte, TypeScript, HTML, and CSS, employing component-based architecture with NgRx and Svelte stores for state management, which improved code reuse and lowered page-load times.
- Build and integrate RESTful services using .NET Core and C#, leveraging Swagger for API documentation, unit tests for validation, and robust exception handling, which enabled reliable data exchange and reduced integration errors.
- Develop and integrate reusable UI components and shared libraries while implementing secure application flows using JWT authentication, claims-based authorization, role-based access control, and protected API communication, which strengthened security compliance and reduced unauthorized-access incidents
- Developed an AI-assisted document-processing microservice using Python, FastAPI, Azure AI Document Intelligence, and Azure AI Language to extract content and redact personally identifiable information, incorporating confidence-based human review and role-based access controls, which speeds up document handling and reduces manual redaction effort

NYSIF | *Software Engineer*

Feb 2025 - May 2026

- Developed backend and server-side applications for policy administration and claims processing with Java 11, Spring Boot, and Kotlin components, using VS Code and PostgreSQL; the services handled statewide insurance workflows and high-volume daily transactions, ensuring reliable processing for all users.
- Designed and implemented a microservices architecture with well-defined service boundaries and API contracts, improving service reliability and enabling independent deployments across multiple teams.
- Built and deployed containerized services on GCP (Cloud Run, GKE), integrating Cloud SQL and Cloud Storage to support secure data and document management, contributing to reduced deployment failures and faster recovery during incidents.
- Built and maintained data pipelines using GCP Pub/Sub to process large-scale data and enable asynchronous processing of policy and claims events, reducing tight coupling and improving system resiliency under load.
- Ensured secure handling of sensitive policy and claims data by implementing role-based access control with Azure AD and enforcing internal data-governance policies, which maintained compliance with regulatory standards and prevented unauthorized-access incidents
- Integrated Spring Security with OAuth2 and JWT to enforce role-based access across internal and partner facing APIs, reducing unauthorized access paths and strengthening compliance with regulatory requirements.
- Designed and operationalized ML-backed services using Vertex AI to support claims risk scoring and predictive analytics, shortening underwriting decision time and improving consistency in risk evaluation.

- Built automated model training and deployment pipelines using Vertex AI Workbench and Vertex Pipelines, enabling repeatable retraining and controlled rollout of ML models into production services.
- Established CI/CD pipelines using GitHub Actions and GCP tooling to automate builds, testing, and deployments, contributing to faster release cycles and lower regression rates.
- Implemented centralized logging and monitoring with GCP Cloud Logging and Cloud Monitoring, improving observability and reducing mean time to detect and resolve production issues.

Capital One | Full Stack Software Engineer

Jan 2024 - Jan 2025

- Designed and developed cloud-native microservices supporting digital banking and payment workflows for 400K+ retail users, ensuring secure transactions, high availability, and consistent system uptime.
- Implemented asynchronous, event-driven data infrastructure using Kafka to process events asynchronously, reducing interservice data exchange latency and improving system reliability under peak traffic.
- Optimized REST and GraphQL APIs through query tuning, indexing, and data-access improvements, reducing average response times by approximately 25% across key customer-facing endpoints.
- Contributed to performance-sensitive data processing tasks by leveraging C++ for optimized computation, improving efficiency in handling high-volume data workflows and reducing processing overhead.
- Built reactive services using Spring WebFlux and implemented circuit breakers, retries, and timeouts with Resilience4J to improve throughput and prevent cascading failures during downstream service disruptions.
- Implemented OAuth 2.0, JWT authentication, and role-based authorization to secure customer and transaction data across internal and customer-facing APIs.
- Containerized and deployed services using Docker and Kubernetes on AWS, supporting scalable releases, service monitoring, and recovery during production incidents.
- Participated in on-call rotations, diagnosing API timeouts, Kafka consumer lag, and downstream service failures through application logs, metrics, and distributed traces to restore production services.

Tata Consultancy Services | Software Developer

May 2021 - Jul 2022

- Modernized a large-scale telecom subscription and billing platform supporting 600K+ accounts by refactoring legacy monolithic components into a microservices-based architecture, improving scalability and long-term maintainability.
- Built and optimized backend services using Java and Spring, applying core concepts including multithreading, collections, exception handling, and functional programming to improve reliability and performance of billing workflows.
- Designed and implemented data access layers using Hibernate and Spring, leveraging HQL, Criteria queries, batch processing, and second-level caching to reduce database load and improve response times by ~30%.
- Developed RESTful APIs with Spring Boot while maintaining SOAP integrations (JAX-WS) for interoperability with legacy systems, ensuring seamless migration without service disruption.
- Implemented structured logging and lightweight health and metrics endpoints, reducing incident triage time by ~25% during high-volume billing and invoicing cycles.
- Built data pipelines to process millions of records daily using Python and FastAPI, optimizing batch processing and reducing manual data handling effort by ~60%.

Vitech Systems Group | Junior Software Engineer

Dec 2020 - Apr 2021

- Contributed to backend services for a policy administration platform used by 50K+ policyholders, building REST APIs with Spring Boot and integrating React-based dashboards.
- Developed and maintained backend components using Java and Spring, supporting core policy workflows and improving API reliability.
- Managed relational databases (e.g., PostgreSQL) by writing queries, updating schemas, and validating data, which improved system accuracy and lowered data-related incidents.
- Investigated and resolved production issues by analyzing logs with the ELK stack and debugging Java services, reducing mean time to resolution and keeping the platform stable for end users.

PROJECTS

AI-Powered Personal Knowledge Engine

Jan 2026

- Built a RAG-based knowledge system to ingest documents, source code, and notes and provide context-grounded answers using vector search and metadata-aware retrieval.
- Developed modular ingestion, chunking, embedding, retrieval, and response-generation services, with source attribution and configurable retrieval strategies.
- Evaluated chunking, ranking, retrieval relevance, latency, and hallucination behavior to improve answer quality and system observability.

EDUCATION

State University of New York at NEW PALTZ | Master of Science

New York, USA May 2024

RVR&JC College Of Engineering | Bachelors of Technology

Andhra Pradesh, India Jun 2021